Indian Academy of Sciences, Bengaluru
Indian National Science Academy, New Delhi
The National Academy of Sciences India, Prayagraj

SUMMER RESEARCH FELLOWSHIPS — 2026

Format for the four-week Report*,^,@

Name of the candidate	:	Tanishka Atul Dorphode
Application Registration no.	:	PHYS1498
Date of joining	:	22 May 2026
Name of the guide	:	Dr. Sandeep Kumar Katarya
Guide's institution	:	Indian Institute of Technology Kanpur
Place of stay during the tenure of the fellowship	:	Hostel provided by Guide: Girls Hostel-1, IIT Kanpur, Kalyanpur, UP Pin - 208016 ~~Own arrangement~~ ~~Other (Specify)~~

Tanishka

Signature of the candidate

Date: 20/06/2026

SandeepKumar

Signature of the guide

Date: 21-6-26

INSPIRE/KVPY FELLOWSHIP (please fill this box)#			
1.	I am currently a recipient of	INSPIRE FELLOWSHIP	☐ Yes / ☑ No
		KVPY FELLOWSHIP	☐ Yes / ☑ No
		If, YES, fill cols. 2, 3 & 4	
2.	**INSPIRE/KVPY** Fellowship is from __-__[month]/__-__[yr] to __-__[month]/__-__[yr]		
3.	I receive a monthly fellowship of Rs. __—__ from INSPIRE/KVPY towards my living expenses		
4.	I also receive towards contingencies a sum of Rs. __—__ per year		
	I affirm that the information given above is correct. Tanishka Signature of the candidate		

IMPORTANT NOTES:

*** The four-week report could be between 300 and 350 words.**

^ This format should be the first page of the report and should be stapled with the main report.

Mandatory to fill this section, this should be filled and signed by you even if you are not an INSPIRE/KVPY Fellow. Otherwise release of fellowship amount will be withheld.

@ The hard copy of the duly signed report should reach the Academy office within 10 days of completing the first month fellowship. If delayed the fellowship amount will not be paid.

(For office use only; do not fill/tear)

I Report	II Report	KVPY	INSPIRE
Candidate's name:		Fellowship amount:	
Student:	Teacher:	Deduction:	
Guide's name:		Amount to be paid:	
KVPY Fellow:	INSPIRE Fellow:	A/c holder's name:	
PFMS Unique Code:		Others	



Summer Research Fellowship Program

Indian Academy of Sciences, Bengaluru
Indian National Science Academy, New Delhi
The National Academy of Sciences, India, Prayagraj

Four-Week Progress Report

Tanishka Atul Doiphode

PHYS1498

Guide: Dr. Sandeep Kumar Kataria

Host Institute: Indian Institute of Technology Kanpur

20 June 2026

Self-Consistent Kinetic Monte Carlo Simulation of Deep Dielectric Charging in Spacecraft Insulation

Introduction

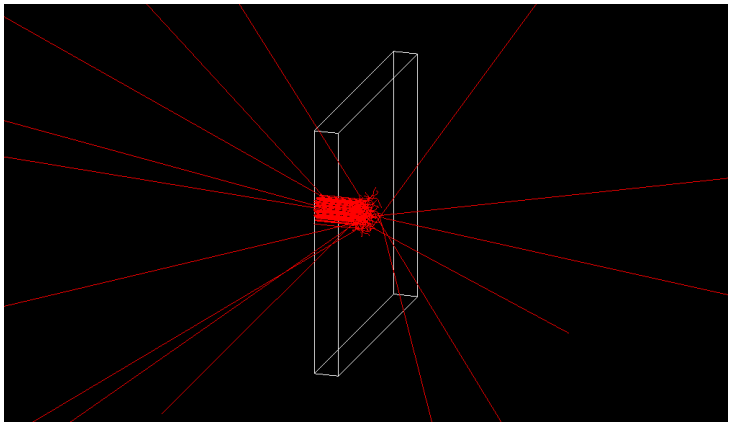
Deep dielectric charging is an important reliability concern for spacecraft operating in radiation-rich environments. Energetic charged particles can penetrate insulating materials and deposit charge within them, leading to electric field buildup and possible electrostatic discharge. Accurate modelling of these effects requires the simulation of particle transport, charge deposition, and secondary particle generation within dielectric materials. In this project, the Geant4 simulation toolkit and the Geant4-based framework *g4chargeit* are being used to investigate charging behaviour in spacecraft insulation materials. The objective is to study the influence of incident particle energy and material properties on charge deposition and secondary particle production.

Methodology

Geant4 examples such as B1 and Microelectronics were studied along with the g4chargeit framework to understand the simulation workflow and output handling. The framework was compiled and modified for preliminary spacecraft charging studies.
A 100 μm thick Kapton slab was implemented as the target geometry. Monoenergetic electrons were used as the incident particles. Event Number, Track ID, and Parent ID were incorporated into the framework output. ROOT-based analysis routines were developed to identify and quantify secondary particle production.

Results

Preliminary simulations were performed for Kapton irradiated by electron beams of different energies. For a representative case involving 100 primary electrons with an incident energy of 100 keV, the simulation produced 319 unique secondary particles, including 318 secondary electrons and 1 photon.
The framework was also modified to record particle lineage information required for secondary emission analysis.



The modified framework successfully simulates particle interactions and secondary particle generation within the Kapton target, as shown in Figure 1. The developed analysis workflow enables identification and quantification of secondary particle production using particle tracking information. These capabilities form the basis for systematic studies of secondary emission behaviour as a function of particle energy and material properties.

Future Work

During the next two weeks, simulations will be carried out over a wider range of incident electron energies and dielectric materials, and the resulting secondary particle yields, energy distributions and depositions resulting in deep dielectric charging will be compared. The final phase of the internship will focus on analysing these trends and evaluating charging behaviour under representative space environment conditions. If time permits, the study will be extended to multilayer insulation (MLI) configurations. Further work beyond the internship may involve validation against published results and incorporation of additional charging physics and simulation framework.

References

1. K. Gandhi et al., *g4chargeit: Geant4-based kinetic Monte Carlo simulations of charging in dielectric materials.*
2. S. Agostinelli et al., *GEANT4 – A Simulation Toolkit*, NIM A, 2003.
3. K. Karafasoulis et al., *GEANT4 simulation study of the response of a miniature radiation detector in Galactic Cosmic Rays and inside a spacecraft*